Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/15/2023
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Mitigating Headward Fluid Shifts with Veno-Constrictive Thigh Cuffs During Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	brandon.r.macias@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Johnson Space Center Cardiovascular and Vision Laboratory		
PI Address 2:	2101 NASA Parkway, HAC/B21N-1207		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/15/2021	**End Date:**	05/14/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Karina Marshall-Goebel and Jonn Foulk moved to new positions and will no longer be participating in the study.		
COI Name (Institution):	Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Lee, Stuart Ph.D. (KBR/NASA Johnson Space Center) Cole, Chris Ph.D. (Clemson University)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This study will determine the ability of venoconstrictive thigh cuff (VTC) application for varying durations to mitigate the weightlessness-induced headward fluid shift during spaceflight for the purpose of evaluating this device as a countermeasure for SANS. The effects of varying durations of continuous VTC usage (~30 minutes, ~3 hours, and ~6 hours) on ocular and cardiovascular variables will be investigated. This proposed study will help determine the potential efficacy of VTC to mitigate the cephalad fluid shift that is hypothesized to be the primary initiating factor to SANS. The VTC countermeasure technique is low-mass, portable, simple to use and may be used for extended durations while working in an operational spaceflight environment. Specific Aim: 1) Determine the efficacy of VTC application to mitigate a spaceflight-induced headward fluid shift. We hypothesize that a VTC countermeasure will temporarily reverse or attenuate spaceflight-induced ocular and cardiovascular changes.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This study will determine the ability of veno-constrictive thigh cuffs (VTC) application for varying durations to mitigate the weightlessness-induced headward fluid shift during spaceflight for the purpose of evaluating this device as a countermeasure for Spaceflight Associated Neuro-ocular Syndrome (SANS). The effects of varying durations of continuous VTC usage (30 minutes, 3 hours, and 6 hours) on ocular and cardiovascular variables will be investigated. This proposed study will help determine the potential efficacy of VTC to mitigate the cephalad fluid shift that is hypothesized to be the primary initiating factor to SANS. The VTC countermeasure technique is low-mass, portable, simple to use and may be used for extended durations while working in an operational spaceflight environment. The investigator team has the required expertise to conduct this research and has experience from a ground study led by this team in 33 subjects who wore these VTC for up to ~2 hours while supine at rest. An updated VTC and microfiber sleeve have been developed and tested in these ground-based studies.
Research Impact/Earth Benefits:	The Flight Thigh Cuff research study will advance NASA's understanding of use of the VTC as a potential countermeasure for Spaceflight Associated Neuro-ocular Syndrome (SANS), an important human health and performance risk. This goal will be accomplished by quantifying vascular and ocular changes before and during spaceflight. Given the unique environment of the International Space Station, commonly used ophthalmic instruments are being implemented in novel operational environments (e.g., optical coherence tomography (OCT) and pneumotonometry). This work will help NASA to evaluate this spaceflight-associated risk more accurately and to refine its data-driven approach to SANS countermeasure development.
Task Progress:	The Feasibility Assessment was completed by the Research and Operations Integration (ROI) team, and the study received Select for Flight. • The Flight Thigh Cuff PI team worked with NASA ROI Element to provide inputs to the Experiment Document, Experiment Summary, and Crew Informed Consent Briefing presentation. • The Flight Thigh Cuff Principal Investigator team worked with NASA ROI, Flight Hardware, and collaborators from the Clemson Textiles Group to implement iterative improvements to the thigh cuff hardware in preparation for the commencement of the study. A flight prototype is targeted for delivery to the Cardiovascular and Vision Laboratory (CVL) by the end of April 2023. • Received IRB approval for the 2023 Institutional Review Board (IRB) Continuing Review. • Received IRB approval for an IRB Protocol and Consent Form modification per request from the European Space Agency Medical Board. • A poster was presented at the 2023 Human Research Program Investigators' Workshop in Galveston, TX.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Abstracts for Journals and Proceedings	Ferguson CR, Laurie SS, Lee SMC, Cole C, Foulk J, Young MH, Macias BR. "Mitigating headward fluid shifts with venoconstrictive thigh cuffs during spaceflight." NASA Human Research Program Investigators' Workshop, Galveston, TX, February 7, 2023 – February 9, 2023. NASA Human Research Program Investigators' Workshop, Galveston, TX, February 7, 2023 – February 9, 2023. , Feb-2023